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Lead Staff Engineer | Full-Stack, AI & Forward Deployed Systems

AI Engineer with over 15 years of experience delivering high-scale architecture. I specialize in designing production-grade **RAG** pipelines and autonomous **AI Agents** using efficient fine-tuning and cost-optimized **LLM** orchestration.

Core Expertise: Generative AI & Agentic Orchestration

- **Production-Grade RAG & Agents:** I specialize in taking LLMs out of the "lab" and into production. I build the enterprise data ingestion and RAG pipelines needed to eliminate hallucinations and make autonomous AI Agents reliable at scale.
- **LLM Selection & Cost Optimization:** I specialize in LLM routing strategies and context-window optimization, leveraging specialized Small Language Models (SLMs) to maintain enterprise-scale performance while minimizing token usage and inference cost.
- **AI Governance & Security:** I focus on AI security, guardrails, compliance, and deterministic orchestration to ensure automated Agentic decision-making is safe for enterprise deployment.

System Architecture & Engineering

- **Modern Full-Stack Leadership:** I build high-performance applications using React, TypeScript, and Node for responsive interfaces, alongside Java and Python for heavy-duty backend orchestration and LLM integration.
- **API & Microservices:** I design clean, decoupled RESTful architectures, GraphQL endpoints, and microservices that make it seamless for cross-functional systems to consume GenAI features.
- **Operational Excellence:** Strong advocate for Test-Driven Development (TDD) for LLM evaluation, CI/CD automation, and robust version control (Git/Perforce) to keep deployment cycles fast and predictable.

Engineering Strategy & Forward Deployment

- **Forward Deployed Architecture:** Act as the primary technical partner for enterprise clients, leading deep-dive workshops to translate complex business needs into production-ready, custom AI systems.
- **Enterprise RAG & Agent Integration:** Partner directly with client engineering teams to securely integrate massive enterprise data cloud streams, optimize model routing, and deploy autonomous AI agents.
- **Client-Facing Systems Delivery:** Lead the end-to-end implementation of high-scale software systems, serving as the technical point of contact to ensure seamless integration and successful delivery for global clients.

Technical Skills

Generative AI & LLM Orchestration:

- **Models:** GPT, Claude, Llama, Gemini, Mistral, Gemma, TinyLlama.
- **Frameworks:** Agent Development Kit (ADK), LangChain, LangGraph, CrewAI, LlamaIndex, Model Context Protocol (MCP)
- **Techniques:** RAG, Context Window Optimization, Prompt Engineering, Memory-Efficient Fine-tuning (LoRA, QLoRA), LLM Cost-Routing, & Agentic Workflow Design.
- **Salesforce AI Ecosystem:** Einstein AI, Prompt Builder, Agentforce / Agent Builder, Data Cloud.

Vector Databases & AI Data Infrastructure:

- **Vector Stores:** Pinecone, ChromaDB, FAISS.
- **Data Ops & Analytics:** NumPy, Pandas.

Web & Full-Stack Architecture:

- **Languages:** Java, Python, JavaScript, TypeScript, Apex.
- **Frontend:** React, Angular, Vue, HTML, CSS, Webpack, Bootstrap, Lightning Web Components (LWC), Salesforce Lightning Design System (SLDS).
- **Backend & API:** Node.js, REST, SOAP, GraphQL, FastAPI, Flask.

Databases & Infrastructure:

- **Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Neo4j.
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, CI/CD (Maven, Ant), Git, Perforce.
- **AI Observability & Testing:** Test-Driven Development (TDD) for web/LLM apps, LangSmith, Phoenix (Arize), Ragas, TruLens, DeepEval, LlamaGuard, Guardrails AI, Weights & Biases.

Development Tools:

- **IDE & AI Tools:** Claude Code, Cursor, Windsurf, VS Code, IntelliJ, PyCharm, Jupyter.

Education

Master's Degree with Business Analytics & Computer Information Systems.

Bachelor's Degree with Mathematics, Physics, Electronics and Micro-Processors.

Certifications & Achievements

Certification (Stanford University) - Machine Learning Specialization

Certification (Sun Microsystems) - Java Programming

Certification (TEK Systems) - Advanced Javascript

Certification (Accelerate) - Aspiring Leaders Program Graduate

Salesforce - AI Agentblazer Champion

Work Experience

Salesforce - San Francisco, CA

Lead Technical Staff Engineer - AI Systems & Web Platforms (January 2012 - Recent)

Architect and deliver the core orchestration infrastructure, full-stack frameworks, and agentic runtimes powering enterprise AI and CRM solutions across the global platform.

- **LLM Orchestration & Gateway Infrastructure:** Architected the high-throughput, low-latency APIs and SDKs bridging web applications with foundational LLMs. Designed secure middleware to handle state management, context window optimization, and prompt chaining across diverse model providers.
- **Autonomous Agentic AI Runtimes:** Led the design and deployment of multi-agent autonomous frameworks capable of complex reasoning, planning, and tool execution. Moved the platform beyond passive chatbots into production-grade systems that execute multi-step business workflows based on natural language intent.
- **Production-Grade RAG Pipelines:** Engineered end-to-end Retrieval-Augmented Generation (RAG) strategies. Spearheaded the integration of real-time, streaming data from Data Cloud and MuleSoft into vectorized contexts, ensuring models are grounded with high-fidelity, deterministic data while strictly respecting enterprise security boundaries.
- **Model Optimization & Fine-Tuning:** Directed technical initiatives for cost and latency optimization, including parameter-efficient fine-tuning (PEFT) for domain-specific CRM tasks. Evaluated and integrated Small Language Models (SLMs) alongside foundational models to optimize inference costs and support localized execution environments.
- **Full-Stack AI Application Engineering:** Developed the web interfaces, core middleware, and extensibility layers (using Node.js, Python, and LWC) that power EinsteinGPT and Agentforce, providing developers with seamless abstractions to build custom AI tools.
- **Scalability & Enterprise Trust:** Built and optimized highly concurrent AI runtimes capable of handling enterprise-scale traffic, ensuring strict compliance, guardrails against hallucinations, and data isolation at every layer of the execution stack.

BlackRock - San Francisco, CA

Senior Software Engineer - Full Stack Web (April 2011 - January 2012)

Led the development and long-term maintenance of mission-critical financial applications, bridging the gap between functional requirements and technical execution.

- **Full-Stack Development:** Built and maintained large-scale, data-driven web applications with a focus on high security and enterprise scalability.
- **System Modernization:** Led the migration of legacy financial applications to modern technology stacks to improve performance and maintainability.
- **API Integration:** Developed robust APIs to bridge multiple financial platforms and third-party application endpoints.
- **Modern UI/UX:** Designed and built responsive, single-page applications (SPAs) focused on improving usability and user efficiency.
- **Cross-Functional Collaboration:** Partnered closely with stakeholders and cross-functional teams throughout the SDLC to ensure technical alignment with business goals.

Yahoo! - Sunnyvale, CA

Software Engineer - Full Stack Web (July 2008 - April 2011)

Managed the development and scaling of software for a global online network, supporting integrated services and a high-traffic advertising platform.

- **Full-Stack Development:** Built and optimized dynamic web applications and backend server logic to handle massive user loads.
- **Systems Ownership:** Collaborated on the full engineering lifecycle, from initial requirement analysis to the design and implementation of core software components.
- **Performance & Security:** Investigated and resolved deep-seated performance bottlenecks and security vulnerabilities within complex web environments.
- **Front-End at Scale:** Developed high-performance user interfaces and custom components using specialized internal frameworks for large-scale web properties.
- **Ad-Tech Specialization:** Contributed to **APT from Yahoo**, one of the industry's largest publisher and advertiser networks, focusing on reliability and feature delivery.
- **Release Engineering:** Streamlined the development process by building and maintaining automated build, package, and deployment scripts.

Prompt Technologies - Cupertino, CA

Software Engineer - Full Stack Web (March 2006 - June 2008)

Led the development and enhancement of mission-critical scheduling platforms for the automotive industry, managing the software from initial requirement analysis through to production.

- **Full-Stack Architecture:** Built scalable web applications using Java-based MVC frameworks (Spring, Struts) to support high-concurrency appointment systems.
- **Rich UI Development:** Developed interactive, "Rich Internet Applications" (RIA) using the Adobe Flex framework, focusing on complex user workflows and real-time interactivity.
- **Enterprise Integration:** Engineered seamless data flows between scheduling platforms and external **Dealership Management Systems (DMS)**, streamlining cross-platform workflows.
- **Performance & Accessibility:** Optimized application performance and ensured cross-browser compatibility to provide a consistent, accessible experience for a diverse user base.
- **Technical Documentation:** Translated business requirements into detailed technical specifications to ensure alignment across the development lifecycle.